

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Eighth Semester B.E. (Credit System) Degree Examinations
 May - June 2023

Duration: 3 Hours

19IS8X83 - SOFTWARE ENGINEERING PRACTICES

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Software Engineering. What are the different types of software applications?	10	L*1	1	1
	b) Explain the incremental development process model with a neat block diagram. List its advantages and disadvantages.	10	L2	2	1
2.	a) Define requirements specification. Explain different ways of writing a system requirement specification.	6	L2	1	1
	b) Explain the different attributes of a good software.	4	L2	1	1
	c) Explain waterfall model of software development process. What are its advantages and disadvantages?	10	L2	2	1
3.	a) What is requirement validation? Explain the checks and techniques followed for process of requirement validation.	10	L2	1	1
	b) Explain the general model of design process.	6	L2	2	1
	c) Summarize the different sectors into which each loop in Boehm's spiral model is split.	4	L2	2	1
Unit – II					
4.	a) Explain the state diagram of a microwave oven.	10	L2	3	1
	b) Explain the activities to be carried out in an object-oriented design process of a system.	10	L2	4	1
5.	a) Illustrate sequence diagram with an example to view patient information.	10	L2	3	1
	b) Explain the principles of agile methods.	5	L2	4	1
	c) Explain the organization of MVC pattern with a neat diagram.	5	L2	4	1
6.	a) Explain transaction processing system with a neat diagram showing software architecture of an ATM system.	6	L2	3	1
	b) Explain generalization with an example.	4	L2	3	1
	c) Summarize the scrum process of agile project management.	10	L2	4	1
Unit – III					
7.	a) What is risk management? Outline the risk management process with a neat diagram and list out the types of risk in a risk checklist.	10	L2	5	1
	b) What are the factors influencing the effectiveness and efficiency of communications in a group?	5	L1	5	1
	c) Explain the factors affecting software pricing.	5	L2	5	1
8.	a) Explain the workflow for a project planning process with a neat diagram.	10	L2	5	1
	b) Explain the three phases of software review process.	6	L2	5	1
	c) What are the benefits of creating a cohesive group?	4	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
